Business case: BaSyx DPP API

Project 6: API for the Digital Product Passport (DPP) in the BaSyx Framework

Customer

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Definition of tasks

> **DIN EN 18222** "Digital product passport – Application programming interfaces (APIs) for product passport lifecycle management and searchability" describes a REST API that is to be implemented in the BaSyx framework as part of this task, both on the backend and frontend sides. The exact task can be found here.

Team 6

Name	e-Mail	Position
Nataliia Chubak	inf24271@lehre.dhbw-stuttgart.de	Project Manager
Luca Schmoll	inf24137@lehre.dhbw-stuttgart.de	Product Manager
Magnus Lörcher	inf24155@lehre.dhbw-stuttgart.de	Product Manager
Manuel Lutz	inf24224@lehre.dhbw-stuttgart.de	Test Manager
Noah Becker	inf24038@lehre.dhbw-stuttgart.de	System Architect
Fabian Steiß	inf24138@lehre.dhbw-stuttgart.de	Technical Writer
Felix Schulz	inf24075@lehre.dhbw-stuttgart.de	UI-Designer

Version Control

Version	Date	Author	Comment
1.0	26.09.2025	Nataliia Chubak	Initial Business Case (BC)
1.1	27.09.2025	Nataliia Chubak	Update BC and financial calculation
1.2	10.10.2025	Luca Schmoll	Convert to Markdown
1.3	02.11.2025	Nataliia Chubak	Add Gantt diagram

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Description and Goals

The goal of the project is to develop a **two-sided (Backend-Frontend) implementation of the REST-API Digital Product Passport (DPP)** in accordance with the **DIN EN 18222 standard** within the **BaSyx framework** to ensure standardized lifecycle management and DPP search.

The relevance of this project lies in **improving the DPP search capability** and **increasing service efficiency**. By providing a ready-made backend and frontend product for DIN EN 18222 standards, the project enables companies to implement the necessary digital solutions more quickly.

Expected Working Hours

Each member of the project has **180 hours** available. The distribution of these hours is detailed below (Table 1).

Activity	Nataliia Chubak	Magnus Lörcher	Luca Schmoll	Fabian Steiß	Noah Becker	Manuel Lutz	Felix Schulz	Total Hours
Product Analysis	20	20	20	0	0	0	0	60
Customer Dialogue	10	5	5	0	0	0	0	20
Leading Project	40	10	10	0	0	0	0	60
Research	15	15	15	15	15	15	15	105
Documentation	25	15	15	30	20	25	25	155
Coding	45	85	85	95	105	75	100	590
Testing	0	0	0	0	0	30	5	35
Meetings	15	15	15	15	15	15	15	105
GitHub Management	5	10	10	20	20	15	15	95
Presentation	5	5	5	5	5	5	5	35
Total	**180**	**180**	**180**	**180**	**180**	**180**	**180**	**1260**

Table 1. Expected working hours

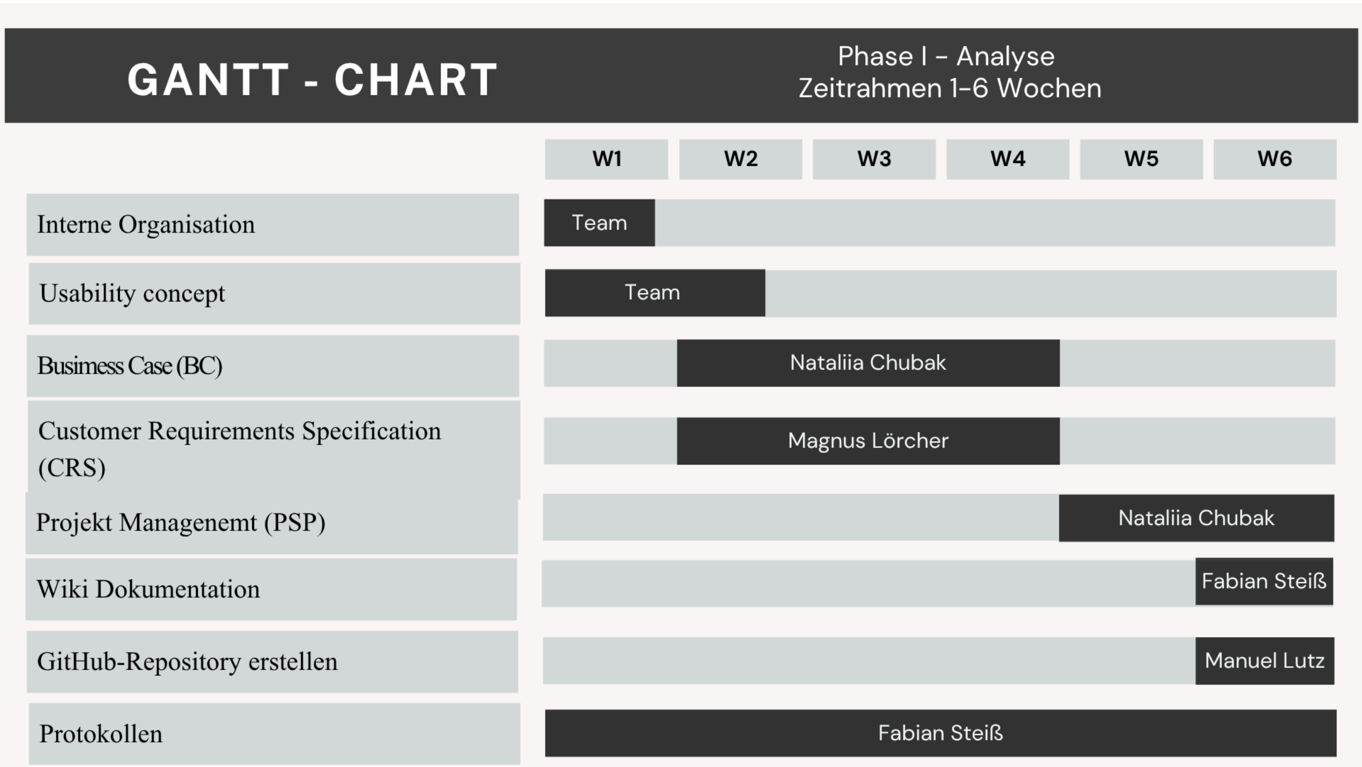
Timeline

The total time frame for this project is **11 weeks** (26.09.2025 – 31.05.2026), with a significant break from 1.12.2025 – 8.03.2026 where team members will be working in their companies.

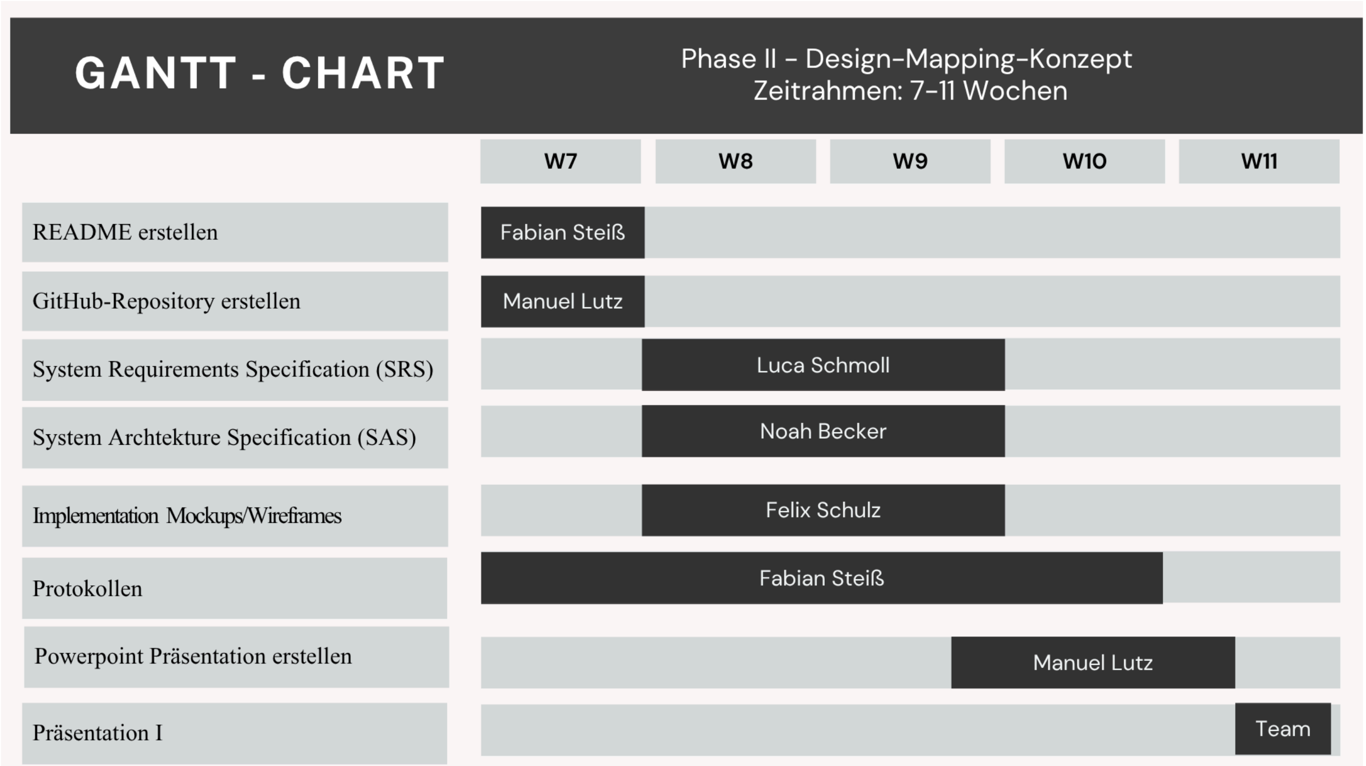
The project milestones are:

- **Planning and Requirements Analysis:** Weeks 1-6
- **Design and Architecture:** Weeks 7-11
- **Development** Weeks 12-17
- **Testing:** Weeks 17-21

Gantt chart 3rd Semester

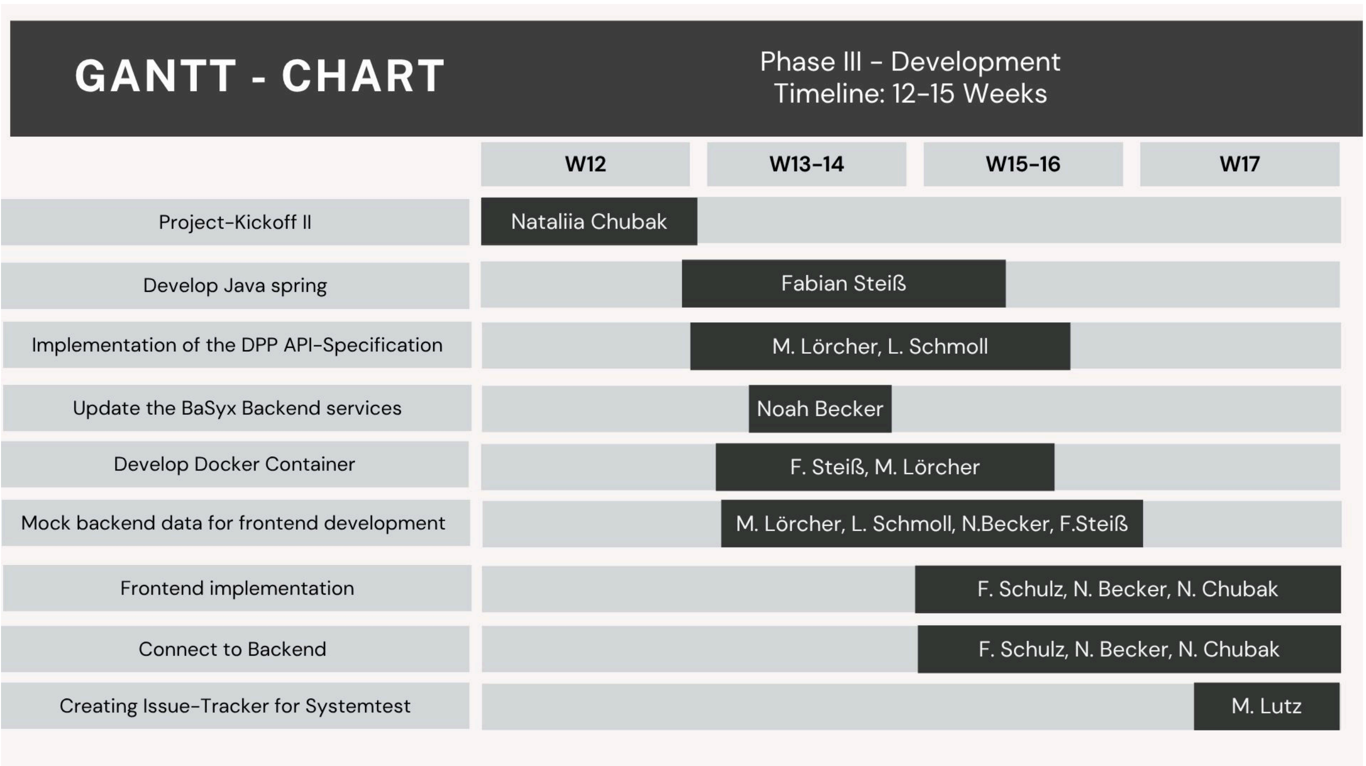


Figur.1 Gantt chart 3rd Semester (Phase I)

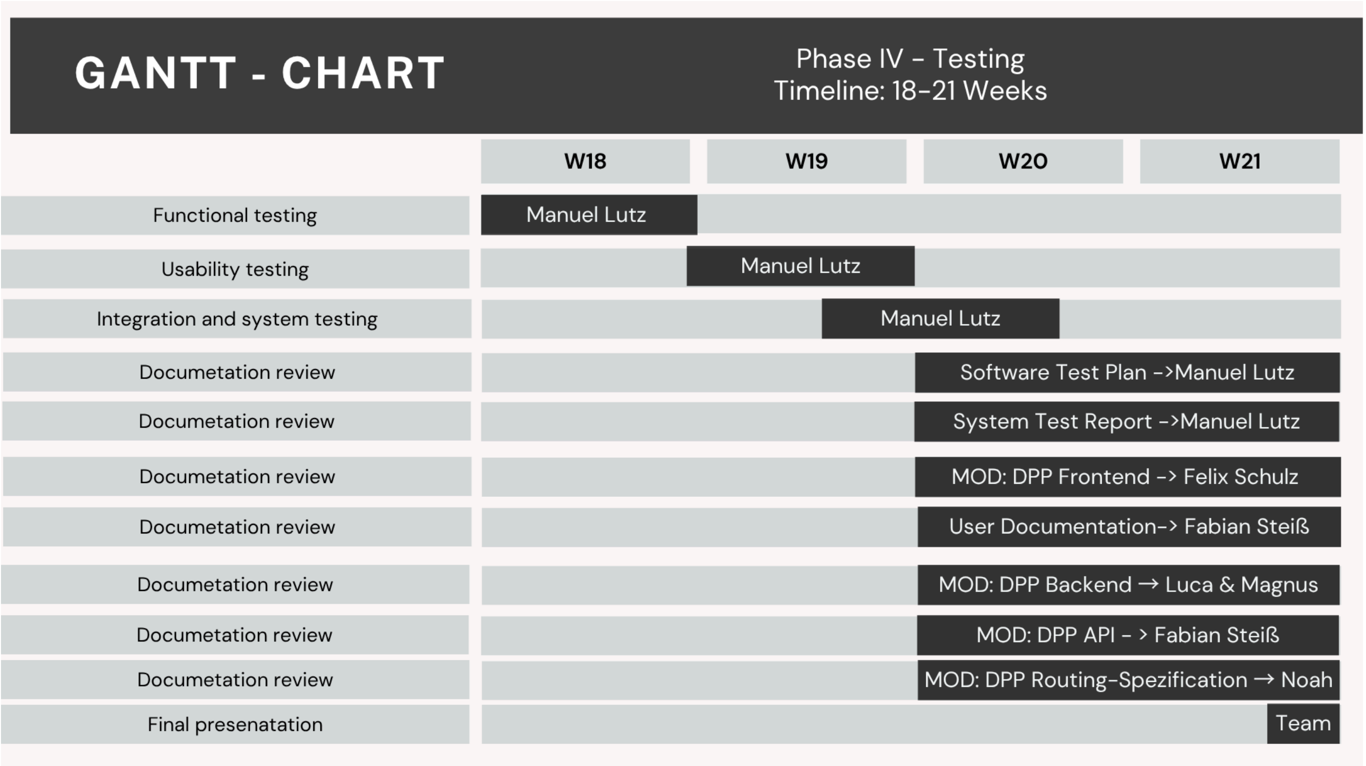


Figur.2 Gantt chart 3rd Semester (Phase II)

Gantt chart 4rd Semester



Figur.3 Gantt chart 4rd Semester (Phase III)



Figur.4 Gantt chart 4d Semester (Phase IV)

Cost Calculation

Calculation of the development cost estimate. The cost estimate is a comprehensive plan of all the company´s expenses for the planned period of production and financial activity.

An investor´s cost estimate includes the following main expenses: • *Base salary*; • *Rent*; • *Components*; • *Electricity*; • *Risk*.

Salary Costs

Based on 180 hours per team member (Table 2).

Position	Salary €/h	Cost Total in €
Project Manager	35	6,300
Product Manager (x2)	32	11,520
Test Manager	30	5,400
System Architect	30	5,400
Technical Writer	28	5,040
UI-Designer	28	5,040
Total Salary Costs	**215**	**38,700**

Table 2. Salary calculation based on hourly wages

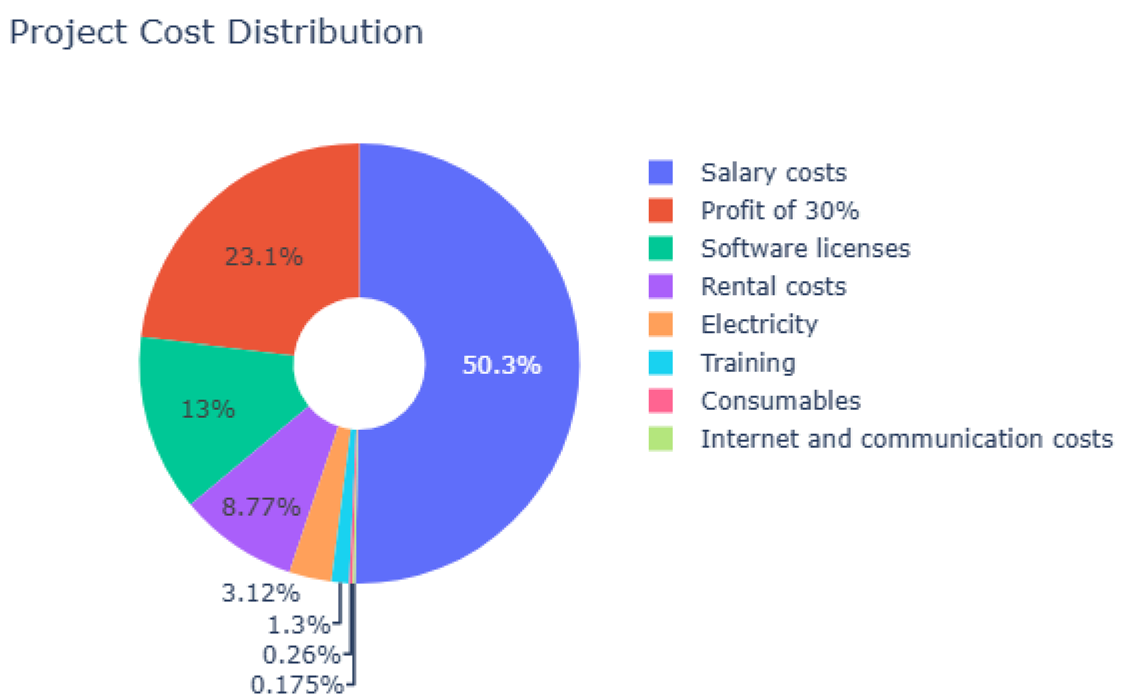
Other Costs

Expenses estimated for the 11-week project duration (Table 3).

Expense Type	Cost Total in € (11 weeks)
Consumables (paper pads, toner, pens)	200
Electricity	2,400
Rental Costs	6,750
Internet services and communication costs	135
Software licenses	10,000
Training	1,000
Total Other Costs	**20,485**

Table 3. Other costs calculation

The estimated total cost of developing the software product is: $$\text{Total Costs} = \text{Salary Costs} + \text{Other Costs}$$ $$\text{Total Costs} = €38,700 + €20,485 = \mathbf{€59,185}$$



Figur.3 Project cost distribution

Risks

The team identified the following key risks:

- *Risks associated with a sudden decline* in the project's operating capacity.

- *Communication risks* (insufficient communication between team members). To solve this problem, it was decided to hold regular meetings, use the GitHub and Jira services.
- **Technical risks** (sudden failure of the Internet connection or power supply for an extended period of time, unplanned unavailability of equipment).
- *Risk of ignoring risks* (insufficiently realistic assessment of scenarios by each member of the test team). To solve this problem, a list of possible risks and possible solutions was created.
- *Schedule risk*. To address this issue, a detailed project work plan was created and progress was systematically monitored.
- *Budget risks*. Exceeding the planned budget due to additional costs.

Offer

This is a commercial offer for the implementation of the DIN EN 18222 standard (Digital Product Passport, DPP) in the Eclipse BaSyx framework. The details and the offer sum of 59 185 € are illustrated in following table. It shows the total costs of the project along with the profit **margin of ~ 30 %** and the offer.

Type of Costs	Costs, €
Salary Costs	38,700
Other Costs	10,485
Total Costs	**59,185**
Profit (30%)	17,755.50
Offer Sum	**76,940.50**

The total offer sum is **€76,940.50**.